Experiment - 02

Aim → Convert the ER diagram into relational schema. write complete MySQL, Create, Table statement with primary key, foreign keys, NOT NULL UNIQUE, ON DELETE CASCADE and set NULL Constraints. Insert sample data and demonstrate referential integrity violations.

Relational Schema

Customer (

Customer ID (PK),
Name,
Email Unique,
Phone no,
DOB

)

Seller (

Seller ID (PK),
Seller Name,
Phone No. Unique,

)

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Address (
    Address ID (PK),
    Costomer ID (FK),
    Pin code,
    state,
    City
)

Category (
    Category ID (PK),
    Category Name unique,
    Description
)

Product (
    Product ID (PK),
    Category ID (FK),
    Price,
    History
)

Fashion (
    Product ID (PK) (FK)
)

Electronic (
    Product ID (PK) FK
)

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Order (
    Order ID (PK),
    Customer ID (FK),
    Order Date,
    Order status
)

Payment (
    Payment ID (PK),
    Order (ID) FK unique,
    Payment Date,
    Amount,
    Status
)

Delivery (
    Delivery ID (PK),
    Order ID (FK) unique,
    Courier Name,
    status,
    Delivery Date
)

Books (
    Product ID (PK) FK,
    Seller ID (FK)
)

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Source Code

```
Create Database ecommerce ;

use ecommerce ;

create table Customer (
        CustomerID int Primary key,
        Name Varchar(50) Not null,
        Email varchar(100) Not null unique
        Phone No. varchar (15) Not NULL,
        DOB Date Not NULL
);

create table Address (
        AddressID int Primary key ;
        CustomerID int Not Null,
        Pincode varchar (10) not null,
        State varchar (50) not null,
        city varchar (50) not null,
        (Foreign key (CustomerID) reference Customer
          (CustomerID) on Delete Casade)
);
```

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```
Create table category (
    category int Primary key,
    category Name varchar (100) not null,
    Description varchar (255)
);

Create table Seller (
    SellerID int Primary key,
    Seller Name varchar (100) not Null,
    Phone No varchar (15) not null unique.
);

create table Product (
    Product ID int Primary key,
    CategoryID int not Null,
    Price Decimal (10, 2) Not null,
    Image varchar (255),
    (Foreign key (category ID) reference
    category (category ID) on Delete cascade)
);
```

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```
Create table orders (
        Order ID int Primary key,
        Customer ID not null,
        Order Date Date not Null;
        order status (varchar(30) not null,
        (Foreign key (customer ID) reference
         customer (customer ID) on Delete Cascade)
);

Create table Delivery (
        Delivery ID int Primary key,
        Order ID int not null unique,
        Courier Name varchar (100) not null,
        status varchar (30)
        Delivery Date Date,
        (Foreign key (order ID) reference orders
        (order ID) on Delete cascade)
);
```

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Inserting Data

```
Insert into Customer (Name, Email, Phone.No, DOB)
values ("Ujjawal", "ujjawal@gmail.com", 7011612101, "2021-01-01")
       ("Krish", "Kri@gmail.com", 1234957896, "2026-04-01");

Insert into category (categoryName, Description)
values ("Electronics", "Electronics Product"),
       ("Books", "Books");

Insert into seller (sellerName, Phone No)
values ("Tech store", "9876543210"),
       ("Book Store", "9999888877");

Insert into Product (category ID), Price, Image)
values (1, 409, "PC.JPG"),
       (2, 569, "Pant.png"),
       (3, 6001, "book.JPG");

Insert into address (Customer ID, Pincode, state, city) values
       (1, "100080", "UP", "Kanpur"),
       (2, "110080", "Delhi", "Saket");
```

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```
Insert into order (customerID, order Date, Order status)
values (1, "2026-08-19", "Placed"),
       (2, "2026-08-19", "Shipped");

Insert into Payment (OrderID, Payment Date, Amount,
Payment Method, status) values.
(1, "2026-08-19", 50000, "UPI", "success"),
(2, "2026-08-19", 19981, "card", "success");

Select * from customer;
Select * from Product;
Select * from orders;
Select * from order Item;

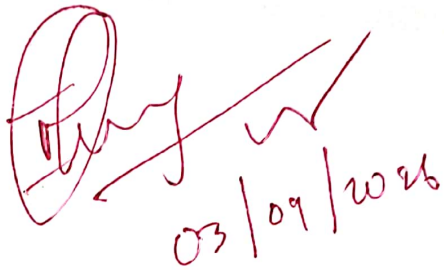
On Delete set NULL
Delete from seller, where sellerID = 2;
Select * from Books;
```

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On Delete Cascade

Delete from customer where Customer ID = 2;

Select * from Orders where customer ID = 2;



03/09/2026

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